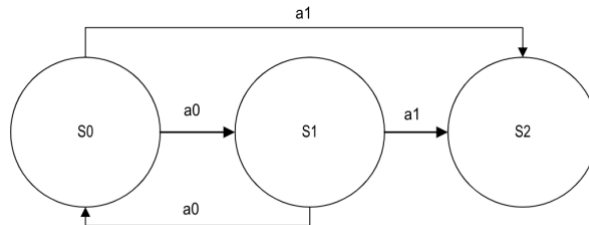


IT5005 Artificial Intelligence
Tutorial 9
SOLUTIONS

1. Consider the Markov Decision Process (MDP) shown below. All transitions are deterministic.



- a. List all the permitted actions in each state.

$$A(S_0) = \{a0, a1\}$$

$$A(S_1) = \{a0, a1\}$$

$$A(S_2) = \emptyset$$

- b. List all the possible deterministic policies.

Four possible policies given the possible actions in S_0 and S_1

$$\pi_0(S_0) = a0$$

$$\pi_0(S_1) = a0$$

$$\pi_1(S_0) = a0$$

$$\pi_1(S_1) = a1$$

$$\pi_2(S_0) = a1$$

$$\pi_2(S_1) = a0$$

$$\pi_3(S_0) = a1$$

$$\pi_3(S_1) = a1$$

- c. Find the value of each state under each policy in part a. The rewards for each action and each state are shown below:

State	a0	a1
S_0	1	1
S_1	2	0
S_2	4	4

We assume that taking an action that is not valid for a particular state has no effect; i.e. we return to the same state. So taking any action in state S_2 will always return to state S_2 . Thus S_2 is a “sink” or “terminal” state, which is quite common since almost all MDPs would end in a terminal state. We assume that $0 \leq \gamma < 1$.

For convenience we state that $V_n \equiv V^{\pi_n}$. We will only present solutions for π_0 and π_1 since the remaining two policies are very similar.

Working out π_0 :

$$\begin{aligned} V_0(S_2) &= 4 + \gamma V_0(S_2) \\ V_0(S_2) &= \frac{4}{1 - \gamma} \end{aligned}$$

$$\begin{aligned} V_0(S_0) &= r(a_0, S_0) + \gamma V_0(S_1) \\ &= 1 + \gamma V_0(S_1) \\ V_0(S_1) &= r(a_0, S_1) + \gamma V_0(S_0) \\ &= 2 + \gamma V_0(S_0) \end{aligned}$$

Substituting $V_0(S_1)$ into $V_0(S_0)$ gives us:

$$\begin{aligned} V_0(S_0) &= 1 + \gamma(2 + \gamma V_0(S_0)) \\ &= 1 + 2\gamma + \gamma^2 V_0(S_0) \end{aligned}$$

Solving for $V_0(S_0)$ we get:

$$V_0(S_0) = \frac{1 + 2\gamma}{1 - \gamma^2}$$

Finally we can get $V_0(S_1)$:

$$\begin{aligned} V_0(S_1) &= 2 + \gamma V_0(S_0) \\ &= 2 + \frac{\gamma(1 + 2\gamma)}{1 - \gamma^2} \end{aligned}$$

For policy π_1 , $V_1(S_2) = V_0(S_2)$

$$V_1(S_2) = \frac{4}{1 - \gamma}$$

$$\begin{aligned} V_1(S_0) &= 1 + \gamma V_1(S_1) \\ V_1(S_1) &= 0 + \gamma V_1(S_2) \\ &= 0 + \gamma \left(\frac{4}{1 - \gamma} \right) \\ &= \frac{4\gamma}{1 - \gamma} \end{aligned}$$

Thus:

$$V_1(S_0) = 1 + \gamma \left(\frac{4\gamma}{1 - \gamma} \right)$$

$$= 1 + \frac{4\gamma^2}{1-\gamma}$$

d. Work out the values for each state for $\gamma = 0, 0.5, 0.9$ for each policy.

Again we will work out only for π_0 and π_1

For π_0 :

$$\begin{aligned} V_0(S_0) &= \frac{1+2\gamma}{1-\gamma^2} \\ V_0(S_1) &= 2 + \frac{\gamma(1+2\gamma)}{1-\gamma^2} \\ V_0(S_2) &= \frac{4}{1-\gamma} \end{aligned}$$

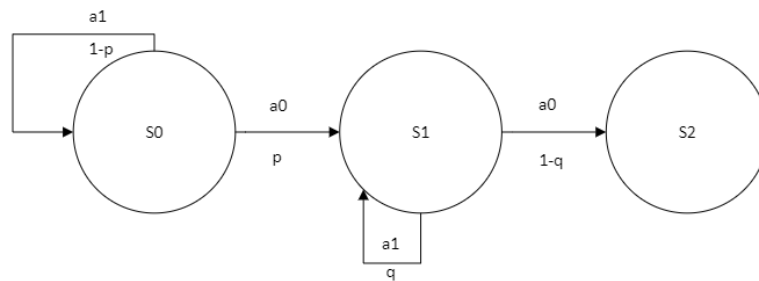
State	$\gamma = 0$	$\gamma = 0.5$	$\gamma = 0.9$
S_0	1	2.67	14.74
S_1	2	3.33	15.26
S_2	4	8	40.0

For π_1 :

$$\begin{aligned} V_1(S_0) &= 1 + \left(\frac{4\gamma^2}{1-\gamma} \right) \\ V_1(S_1) &= \frac{4\gamma}{1-\gamma} \\ V_1(S_2) &= \frac{4}{1-\gamma} \end{aligned}$$

State	$\gamma = 0$	$\gamma = 0.5$	$\gamma = 0.9$
S_0	1	3	33.4
S_1	0	4	36
S_2	4	8	40

2. Consider now a probabilistic MDP as shown below. The actions are shown above the edges representing the actions and transitions, while the probabilities of taking those actions are shown below:



For example in state S_0 there is a probability p of taking action $a0$ and moving to state S_1 , and a probability $1 - p$ of taking action $a1$ and returning to state S_0 . As before S_2 is a terminal state and taking any action there will always return you back to S_2 .

Find the value of each state given the following rewards:

State	a0	a1
S_0	2	3
S_1	3	1
S_2	2	2

For $V(S_2)$:

$$\begin{aligned}
 V(S_2) &= 2 + \gamma V(S_2) \\
 &= \frac{2}{1 - \gamma}
 \end{aligned}$$

For $V(S_0)$:

$$\begin{aligned}
 V(S_0) &= p(2 + \gamma V(S_1)) + (1 - p)(3 + \gamma V(S_0)) \\
 &= \frac{3 + p(\gamma V(S_1) - 1)}{1 - \gamma(1 - p)}
 \end{aligned}$$

For $V(S_1)$:

$$\begin{aligned}
 V(S_1) &= q(1 + \gamma V(S_1)) + (1 - q)(3 + \gamma V(S_2)) \\
 &= q(1 + \gamma V(S_1)) + (1 - q)\left(3 + \frac{2}{1 - \gamma}\right)
 \end{aligned}$$

$$\begin{aligned}
&= q(1 + \gamma V(S_1)) + (1 - q)\left(\frac{5 - 3\gamma}{1 - \gamma}\right) \\
&= q(1 + \gamma V(S_1)) + \frac{(1 - q)(5 - 3\gamma)}{1 - \gamma} \\
&= q + q\gamma V(S_1) + \frac{(1 - q)(5 - 3\gamma)}{1 - \gamma} \\
&= q\gamma V(S_1) + \frac{q(1 - \gamma) + (1 - q)(5 - 3\gamma)}{1 - \gamma} \\
V(S_1) &= \frac{q(1 - \gamma) + (1 - q)(5 - 3\gamma)}{(1 - q\gamma)(1 - \gamma)}
\end{aligned}$$

We can substitute this back into $V(S_0)$ above but this step is omitted. It's very tedious but trivial.

3. One urn contains 10 red balls and 25 green balls, and a second urn contains 22 red balls and 15 green balls. A ball is chosen as follows: First an urn is selected by tossing a loaded coin with probability 0.4 of landing heads up and probability of 0.6 of landing tails up. If the coin lands heads up, the first urn is chosen; otherwise, the second urn is chosen. Then a ball is picked at random from the chosen urn.

Write your answers correct to three significant figures.

- (a) What is the probability that the chosen ball is green?
(b) If the chosen ball is green, what is the probability that it was picked from the first urn?

Answers:

(a) $\frac{4}{10} \cdot \frac{25}{35} + \frac{6}{10} \cdot \frac{15}{37} = 52.9\%.$

Probability that the chosen ball is green is **52.9%**.

- (b) Let G be the event that the chosen ball is green, U_1 the event that the ball came from the first urn, and U_2 the event that the ball came from the second urn.

$$P(U_1) = 0.4, P(U_2) = 0.6, P(G|U_1) = \frac{25}{35}, P(G|U_2) = \frac{15}{37}.$$

By Bayes' Theorem,

$$\begin{aligned}
P(U_1|G) &= \frac{P(G|U_1) \cdot P(U_1)}{P(G|U_1) \cdot P(U_1) + P(G|U_2) \cdot P(U_2)} \\
&= \frac{\left(\frac{25}{35}\right) \cdot 0.4}{\left(\frac{25}{35}\right) \cdot 0.4 + \left(\frac{15}{37}\right) \cdot 0.6} = \frac{74}{137} = 54.0\%
\end{aligned}$$

Therefore, the probability that the green ball is chosen from the first urn is **54.0%**.